



▲ LIGHT AND COLOR

Even in quite shallow water, everything appears very blue. This is because all the other colors that make up white sunlight are absorbed by the water. Red light is absorbed first, followed by yellow, then green and violet, until only blue light is left. At greater depths even this fades, leaving total darkness. This means that marine organisms that need light to make food, such as seaweeds and microscopic plankton, can only grow and multiply near the sunlit surface.



◀ LIGHT ZONES

The filtering of light by ocean water creates three main light zones. In the sunlit zone, there is enough light for seaweeds and plankton to flourish, and support other animals. Below 650 ft (200 m) is the twilight zone, where there is only dim blue light. Fewer animals live here, although many move down from the sunlit zone during the day. About 3,300 ft (1,000 m) below lies the dark zone, where there is no light at all except the strange luminous glow produced by many of the animals that live in the deep ocean.



PRESSURE

The enormous weight of ocean water exerts crushing pressure at depth. Humans are adapted to cope with atmospheric pressure, known as 1 bar. Just 33 ft (10 m) below the ocean surface, the pressure doubles to 2 bars, and at 65 ft (20 m) it increases to 3 bars. On the ocean floor, 10,000 ft (3,000 m) below the surface, it increases to some 400 times normal atmospheric pressure. This means that divers who work at depths below 165 ft (50 m) must wear special pressure-proof suits like this one. Manned submersibles designed to reach the dark zone have to be extremely strong, with the crew traveling inside a metal sphere that is designed to resist the colossal pressures that exist near the ocean floor.



▲ SOUND

Sound travels through water five times faster than through air, and this enables whales, for example, to communicate over great distances. Sound transmission is most efficient at depths of around 3,300 ft (1,000 m) in a region called the SOFAR (Sound Fixing and Ranging) channel. Any sound generated within this zone cannot escape, but is reflected back into and along it. This focussing effect enables sounds to travel astonishing distances of up to 15,500 miles (25,000 km).